The complete dS_3 crossing kernel on its heavy-mass peak: phase normalization and an explicit remainder

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Abstract

After Wilson-duality stabilization, the complete scalar crossing kernel admits a controlled expansion on the exact collision surface. We retain its physical CG-period phase, bound the large-parameter companion series as a whole, and obtain the leading logarithm, constant, and inverse-mass correction with a compact-transfer uniform remainder. An explicit finite-mass bound is also given. This is a peak-surface result for a specified $G_N = 0$ model; it neither establishes a distributional delta limit nor interchanges a physical mass integral with the heavy-observer limit.

1 Statement, labels and physical scope

We continue with radius one, $G_N = 0$, proper $SO^+(3,1)$, BD scalar observers and KG matter, a fixed positive observer-label lower bound, and half-period regulated creation blocks. The normalized scalar-to-scalar element U_{00} retains the physical CG-period phase. It is not a separate unitary scalar transform. The external labels are (a, q, c, p) and internal labels $(b, d) > 0$. Here M is a principal-series label; the static energy is $\sqrt{1+M^2}$. No energy Jacobian or mass integration is being suppressed.

On the exact collision surface choose

$$c = M, \quad b = M + u, \quad a = M + u + v, \quad d = M + v, \qquad |u| + p \le 1, \quad |v| + q \le 1. \tag{1}$$

Define $A_j = 1/2 + i\alpha_j$ by

$$(\alpha_1, \alpha_2, \alpha_3, \alpha_4) = \tfrac{1}{2}(v+q,\ p-u,\ -u-p,\ v-q).$$

For fixed positive p, q and fixed u, v in this window, we obtain

$$\boxed{U_{00} = \frac{\log M}{\pi} + \frac{2\psi(1) - \Re\sum_j \psi(A_j)}{2\pi} + \frac{u+v}{2\pi M} + O\!\left(\frac{\log M}{M^2}\right).} \tag{2}$$

The remainder is uniform on compact admissible transfer sets with $p, q > 0$. All observer labels exceed the fixed lower bound once M is sufficiently large. The proof below is a model calculation built from the known LPRS kernel and standard special-function identities [1, 2, 3]. No originality or full-duality claim is made.

A logarithmically high ridge by itself determines neither the width nor the integrated weight of a kernel. Equation (2) is not used as a delta replacement in any observable.

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2 The physical phase and the two series

Write $h = (1+ib)/2$, $h' = (1+id)/2$ and $h_i = (1+i\nu_i)/2$. The preceding Wilson reflection identity gives

$$U_{00} = C_+ XY + \widehat{C}_- \,\overline{XY}, \qquad X = \Omega(h,h'), \quad Y = \Omega(1-h,h'), \quad |\widehat{C}_-| = |C_+|.$$

In the present CG-period convention the stronger exact identity is

$$\widehat{C}_- = \overline{C_+}, \qquad U_{00} = 2\Re(C_+ XY). \tag{3}$$

This is an analytic phase identity, not a numerical real projection. For reproducibility, the period for three all-conjugated legs is

$$N(x,y,z) = (4\pi)^{3/2} 4^{-3/2-iS/2} \frac{\Gamma(\frac{1}{2} - \frac{iS}{2}) \prod_{\nu=x,y,z} \Gamma(\frac{1}{2} - \frac{i(S-2\nu)}{2})}{\prod_{\nu=x,y,z} \Gamma(1-i\nu)}, \quad S = x+y+z.$$

The physical phase is that of $N(a,q,b)N(c,p,-b)/[N(c,q,d)N(a,p,-d)]$. Substitute this and the explicit four-Gamma reflection ratios into $\widehat{C}_- C_+$. All powers of four cancel in phase; the remaining Gamma arguments occur in conjugate pairs with equal integer weights. Their product is positive. Combined with the previously proved modulus identity, this proves (3). The full argument/weight list is saved with an exact symbolic check for six independent real labels.

On (1), LPRS (3.38) separates X into

$$X = c_A F + c_B G, \quad F = {}_4F_3(A_1, A_2, A_3, A_4; 1+id, 1, 1-ib; 1),$$

$$G = {}_4F_3(A_1+ib, \ldots, A_4+ib; 1+i(b+d), 1+ib, 1+ib; 1),$$

$$c_A = \frac{\Gamma(ib)\Gamma(1+ib)\Gamma(A_3)\Gamma(A_4)}{\Gamma(A_3+ib)\Gamma(A_4+ib)},$$
$$c_B = \frac{\Gamma(-ib)\Gamma(1+id)\Gamma(A_1+ib)\Gamma(A_2+ib)}{\Gamma(A_1)\Gamma(A_2)\Gamma(1+i(b+d))}.$$

Set $D_0 = C_+ c_A$ and $E_0 = C_+ c_B$. Gamma modulus identities imply

$$D_0 = \frac{e^{i\theta(M)}}{4\pi}, \qquad \theta(M) = \frac{-p^2 v + q^2 u + u^2 v - uv^2 + u - v}{4M^2} + O(M^{-3}). \tag{4}$$

For the modulus, the $\sinh(\pi b)$ in $|C_+|$ cancels that in $|\Gamma(ib)\Gamma(1+ib)|$; the four remaining cosh factors cancel. For the phase, insert the displayed period and apply the shifted Stirling expansion to each Gamma argument $\pm iM + \eta$. The small-argument Gamma phases cancel in conjugate pairs. The $M\log M$, M, $\log M$, constant and M^{-1} terms all cancel; the Bernoulli-polynomial term of order M^{-2} gives (4). The asymptotic continuous phase is chosen to tend to zero. The shifts range over a compact set, away from the Stirling branch cut, so the remainder is uniform.

3 Bounding the entire large-parameter companion

It is not sufficient to keep just the first term of G. Put $A_* = \max_j |\alpha_j| \le 1/2$ and write $G = \sum_{n\ge 0} g_n$. For $k = n+1$, since $d > 0$,

$$\left|\frac{g_{n+1}}{g_n}\right| \le \frac{[k^2 + (b + A_*)^2]^2}{k[k^2 + b^2]^{3/2}} = \sqrt{1 + \frac{b^2}{k^2}} \left(1 + \frac{2bA_* + A_*^2}{k^2 + b^2}\right)^2. \tag{5}$$

The upper real parts $1/2$ were enlarged to one in this bound. Using $\log(1 + t) \le t$, the sinh product [4], and $\sum_{k\ge 1}(k^2 + b^2)^{-1} \le \pi/(2b)$ gives, for every n,

$$|g_n| \le B_b, \qquad B_b = \sqrt{\frac{\sinh \pi b}{\pi b}} \exp\left(2\pi A_* + \frac{\pi A_*^2}{b}\right). \tag{6}$$

To sum this bound, use the preceding tail-certificate polynomial with upper imaginary parts $b + \alpha_j$ and lower parts $(b + d, b, b)$:

$$\mathcal{A}(x) = \prod_j [(x + \tfrac{1}{2})^2 + (b + \alpha_j)^2], \quad \mathcal{B}(x) = (x + 1)^2[(x + 1)^2 + (b + d)^2][(x + 1)^2 + b^2]^2,$$

$$P(x) = \mathcal{B}(x)(x + 1)^3 - \mathcal{A}(x)(x + 2)^3 = x^{10} + \sum_{j=0}^{9} p_j x^j.$$

Let $K_* = \max_{j<10} |p_j|$ and $N_* = 2 + \lceil K_* \rceil$. Every normalized derivative of P is monic with lower coefficients bounded by K_*. For $x > 1 + K_*$ its lower powers have total absolute value smaller than the leading power, by the geometric-series bound. Thus every derivative of P is positive at N_*, and $P(N_* + y)$ has positive coefficients. Consequently

$$|g_{n+1}/g_n| \le ((n + 1)/(n + 2))^{3/2} \quad (n \ge N_*).$$

Split the sum before N_* and use the integral-test tail:

$$|G| \le \sum_n |g_n| \le (3N_* + 3)B_b =: B_G. \tag{7}$$

N_* is an auxiliary analytic threshold, not a loop length used by our numerical calculation.

On a compact transfer set $K_* = O(M^8)$. Gamma asymptotics give $|E_0| = O(M^{-1/2}e^{-\pi M})$ and $B_b = O(M^{-1/2}e^{\pi M/2})$. Hence

$$|E_0 G| \le |E_0| B_G = O(M^8 e^{-\pi M/2}). \tag{8}$$

This deliberately loose exponential estimate is sufficient for (2). It retains the whole companion series rather than assuming it is close to one.

4 Finite-mass bound and proof of the expansion

The small-parameter series F shares the coefficients of the collision digamma series for Y. Let $m = \min(b, d)$ and

$$q_0 = \psi(1 - ib) + \psi(1 + id) + 2\psi(1) - \sum_j \psi(A_j),$$

$$B_M = 2 + \operatorname{arsinh} b + \operatorname{arsinh} d + 4 \log 2 \sum_j |A_j - 1|.$$

The already established coefficient comparison in the window $|\alpha_j| \le 1/2$ gives

$$|F - 1| \le R_F = \frac{\zeta(2) - 1}{1 + m^2}, \qquad |Y - q_0| \le R_Y = \frac{(\zeta(2) - 1)B_M}{1 + m^2}. \tag{9}$$

For example the term estimate follows from $(x^2 - x + 1/2)(x + 1) \le x^3$ for $x \ge 1$; summing the resulting $(n + 1)^{-2}$ bound gives the first inequality. The digamma difference-series bound supplies B_M for the second. These are exact-arithmetic analytic bounds.

Combine (3) with $X = c_A F + c_B G$:

$$U_{00} = \frac{1}{2\pi} \Re(e^{i\theta} FY) + 2\Re(E_0 GY).$$

For the exact finite-mass quantities, an explicit absolute estimate is

$$\left| U_{00} - \frac{\Re q_0}{2\pi} \right| \le \frac{R_Y}{2\pi} + \left[\frac{R_F + |e^{i\theta} - 1|}{2\pi} + 2|E_0| B_G \right] (|q_0| + R_Y). \tag{10}$$

The implementation evaluates $|e^{i\theta} - 1| = |4\pi D_0 - 1|$ without a phase-unwrapping assumption. Its numerical evaluation, unlike the analytic inequality, is not interval arithmetic.

Uniformly on compact admissible transfers, the estimates are

$$q_0 = O(\log M), \quad R_F = O(M^{-2}), \quad R_Y = O(\log M / M^2), \quad e^{i\theta} - 1 = O(M^{-2}).$$

Equation (8) then reduces (10) to

$$U_{00} = \frac{\Re q_0}{2\pi} + O(\log M / M^2).$$

Finally the standard digamma expansion gives

$$\Re[\psi(1 - ib) + \psi(1 + id)] = \log b + \log d + O(M^{-2}) = 2 \log M + \frac{u + v}{M} + O(M^{-2}).$$

This proves (2), including the specified phase convention.

The proof has not controlled a neighborhood transverse to $a + c = b + d$, nor an integral over that transverse direction. A delta limit would require its width, oscillatory tails, normalization against test functions and the actual physical spectral weights. The next calculation must supply those ingredients before using (2) inside a Bose four-point function.

5 Numerical record and limitations

Two transfer choices are tested: $(u, v, p, q) = (0.3, -0.2, 0.6, 0.6)$ and $(0.2, 0.1, 0.7, 0.8)$. For each, six masses $M = 8, 16, 32, 64, 128, 256$ are evaluated with the reflected complete kernel, 128 terms and 100 decimal digits. All twelve observed discrepancies from $\Re q_0/(2\pi)$ satisfy (10), allowing separately for the saved series tail.

For the first transfer choice:

M	observed discrepancy	explicit bound (10)
8	$2.81\,10^{-3}$	$1.13\,10^{4}$
16	$8.55\,10^{-4}$	5.15
32	$2.53\,10^{-4}$	$3.16\,10^{-3}$
64	$7.29\,10^{-5}$	$8.64\,10^{-4}$
128	$2.07\,10^{-5}$	$2.35\,10^{-4}$
256	$5.77\,10^{-6}$	$6.36\,10^{-5}$

The very loose low-mass bounds are retained. The scaled residual $M^2|U - U_{\text{expansion}}|/\log M$ decreases from approximately 0.0941 to 0.0711 for this sample. Also $M^2\theta$ approaches the analytic coefficient 0.1625; the second sample approaches 0.04525. These observations check the formulas but do not replace the proof.

The first independent arithmetic audit failed because it imported a roughly 70-digit value of π while requesting 85–90-digit comparisons. That audit source and failed output remain unchanged. A separate replay computes π from the Machin formula at 180 digits, with an alternating-series truncation bound below 10^{-160}. With unchanged scientific data and unchanged comparison tolerances, all 191 replay checks pass. The checks include exact rational positivity at the auxiliary threshold, finite-bound reconstruction, complex residuals and Gamma-phase conjugate pairing. This still does not certify arithmetic rounding.

All code, inputs, versions, full complex outputs, failed checks and repairs are retained with hashes. The earlier 100-digit parameter-derivative job remains running at this checkpoint and is not counted as passed. Final project registration and freezing of the recent stages remain pending.

The peak-surface result advances the complete-kernel analysis, while the off-surface uniform limit, physical wave-packet integration, Bose free-observer limit, full Fock domains, finite-G_N dynamics, and microscopic holographic dictionary remain open.

References

[1] J. Liu et al., *d-dimensional SYK, AdS Loops, and 6j Symbols*, arXiv:1808.00612v3, Eqs. (3.35)–(3.38).

[2] W. Groenevelt, *The Wilson function transform*, arXiv:math/0306424v1, Sec. 3 and Proposition 4.4.

[3] NIST DLMF, Sec. 5.11, shifted Stirling and digamma asymptotics.

[4] NIST DLMF, Sec. 4.36.1, sinh infinite product. Consulted 31 October 2026.